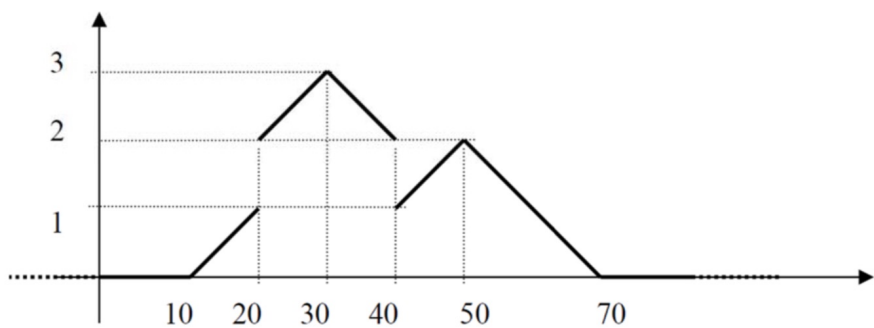


Exercise 1

It is given the signal $s(t)$ represented in the figure below, in which the times (on the x-axis) are expressed in microseconds:



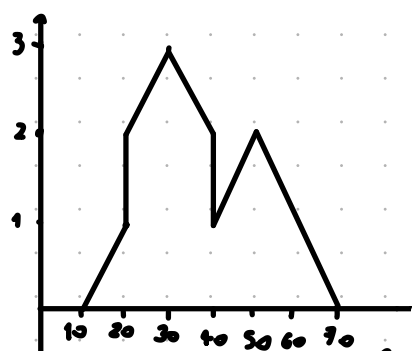
Write the Fourier transform of the $s(t)$ signal.

Calculate the energy of the portion of the signal between between 0 and 40 μs .

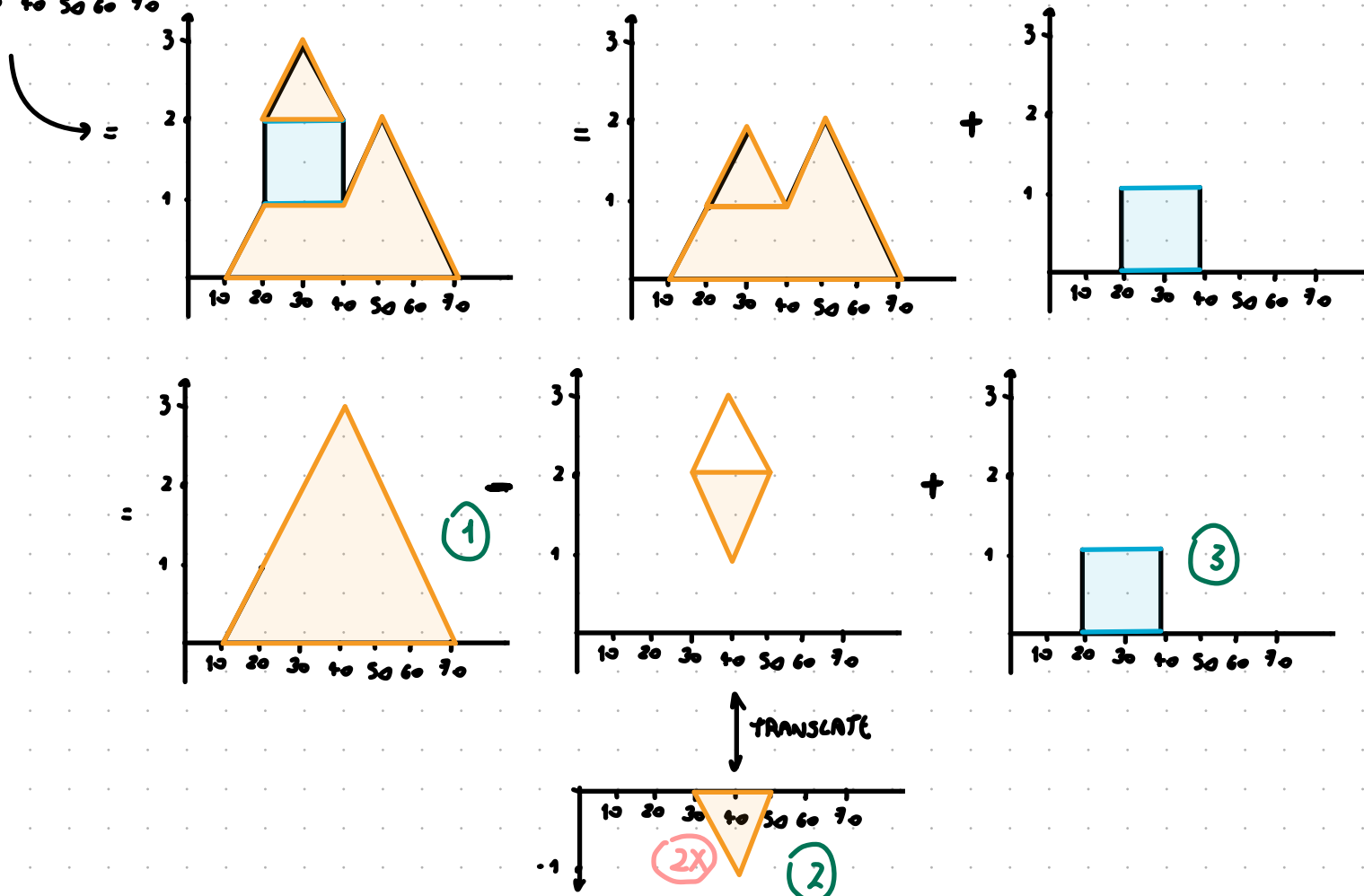
Without performing the relevant calculation, can you estimate whether the energy of the remaining portion (from 40 μs onward) is greater or less than the energy in the previous tract?

How large is the coefficient a_0 of the Fourier series associated with the periodic signal $s_p(t)$ that is obtained by replicating infinite times the part of $s(t)$ between between 0 and 80 μs ?

LOOK DOWN



② CALCULATE FOURIER TRANSFORM
 • SIGNAL IS APERIODIC
 • IT IS POSSIBLE TO DECOMPOSE A SIGNAL IN SIMPLER ONES (TRIANG AND RECT)



TRIANGLES: $S(f) = A \cdot \left(\frac{\text{BASE}}{2}\right) \cdot \text{sinc}^2\left(f \frac{\text{BASE}}{2}\right) \cdot e^{-j2\pi f \cdot \text{PEAK_DELAY}}$
 RECTANGLES: $S(f) = A \cdot (\text{BASE}) \cdot \text{sinc}(f \text{ BASE}) e^{-j2\pi f \cdot \text{PEAK_DELAY}}$

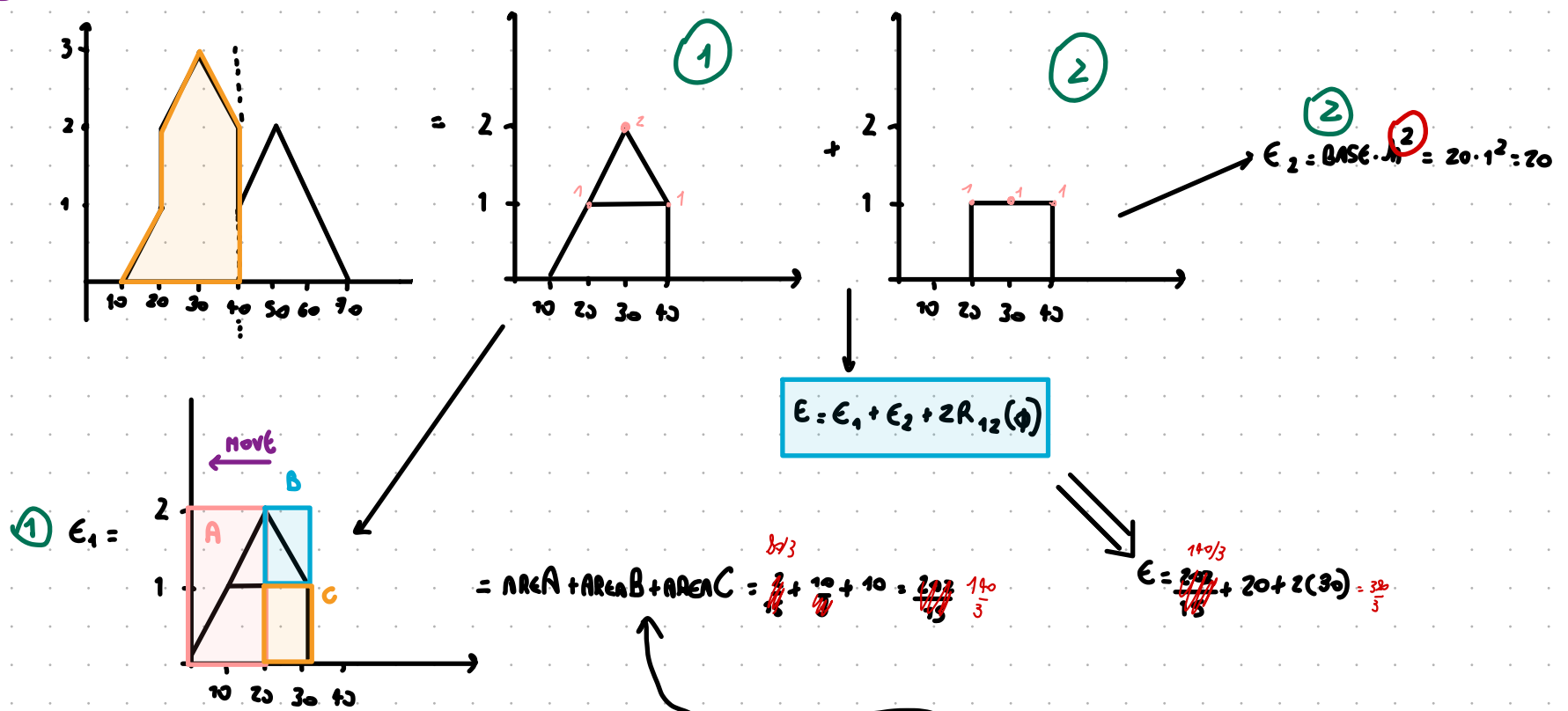
$$\textcircled{1} S_1(f) = 3 \cdot \frac{60}{2} \cdot \text{sinc}^2\left(f \frac{60}{2}\right) \cdot e^{-j2\pi f \cdot 40} = 3 \cdot 30 \text{sinc}^2(f 30) e^{-j2\pi f 40}$$

$$\textcircled{2} S_2(f) = 2 \cdot -1 \cdot \frac{20}{2} \text{sinc}^2\left(f \frac{20}{2}\right) \cdot e^{-j2\pi f \cdot 40} = 2 \cdot (-10 \text{sinc}^2(f 10) e^{-j2\pi f 40}) = -20 \text{sinc}^2(f 10) e^{-j2\pi f 40}$$

$$\textcircled{3} S_3(f) = 1 \cdot 20 \text{sinc}(f 20) \cdot e^{-j2\pi f \cdot 30}$$

$$\hookrightarrow S(f) = S_1(f) + S_2(f) + S_3(f)$$

⑥ CALCULATE ENERGY OF SIGNAL BETWEEN 0 AND 40 MS



① $y = \omega x^2 \rightarrow 2 = \omega 20^2 \rightarrow \omega = \frac{1}{200}$

$\text{AREA A} = \int_0^{20} \omega t^2 dt = \frac{1}{200} \int_0^{20} t^2 dt = \frac{1}{200} \left[\frac{t^3}{3} \right]_0^{20} = \frac{1}{200} \left(\frac{20^3}{3} - 0 \right) = \frac{200}{3}$

② $y = \omega x^2 \rightarrow 1 = \omega (-10)^2 \rightarrow \omega = \frac{3}{100}$

$\text{AREA B} = \int_{-10}^0 \omega t^2 dt = \frac{3}{100} \int_{-10}^0 t^2 dt = \frac{3}{100} \left[\frac{t^3}{3} \right]_{-10}^0 = \frac{3}{100} \left(0 - \frac{(-10)^3}{3} \right) = \frac{10}{3}$

③ $\text{AREA C} = 10 \cdot 1 = 10$

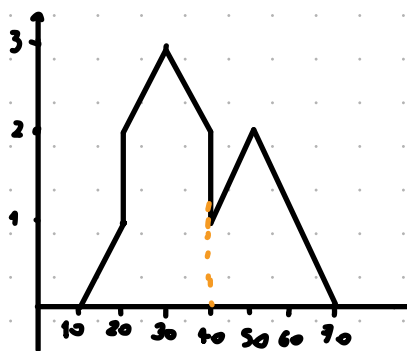
$R_{12}(\phi) = \int s_1(t) s_2(t) dt =$

$R_{12}(\phi) = A_{\text{TOT}} = A_{\text{RECT}} + A_{\text{TRIANG}} = \text{BASE} \cdot h + \frac{\text{BASE} \cdot h}{2} = 20 \cdot 1 + \frac{20 \cdot 1}{2} = 30$

Since is a product and in this area s_2 does not exist the result is 0

LOOK DOWN

⑦c ESTIMATE ENERGY FOR REMAINING GRAPH [40; 70]



FROM THE GRAPH SIZE, KNOW $E(10; 40) > E(40; 70)$

How large is the coefficient a_0 of the Fourier series associated with the periodic signal $s_p(t)$ that is obtained by replicating infinite times the part of $s(t)$ between between 0 and $80 \mu s$?

- $T_0 = 80 \mu s$
- $C_0 = ?$
- GENERIC FORMULA:

THIS IS THE AREA (SINCE IT IS AN INTEGRAL) OF THE SIGNAL $s(t)$.

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t) e^{-j2\pi n t / T_0} dt \xrightarrow{n=0} C_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t) dt = \frac{1}{80} \left[2 \left(\frac{20 \cdot 2}{2} + \frac{(2+1) \cdot 10}{2} \right) + 20 \cdot 1 \right] = \frac{1}{80} \cdot 90 = \frac{9}{8}$$

MIRRORED

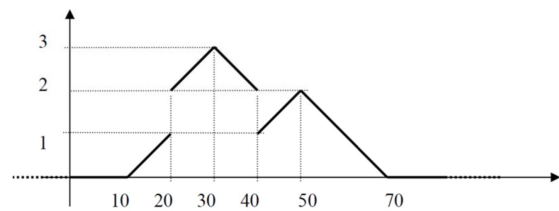
BASE TRIANGLE (a1)
BASE TRIANGLE (a2)
RECTANGLE (a3)



LOOK DOWN

BETTER ↓

It is given the signal $s(t)$ represented in the figure below, in which the times (on the x-axis) are expressed in microseconds:



Write the Fourier transform of the $s(t)$ signal.

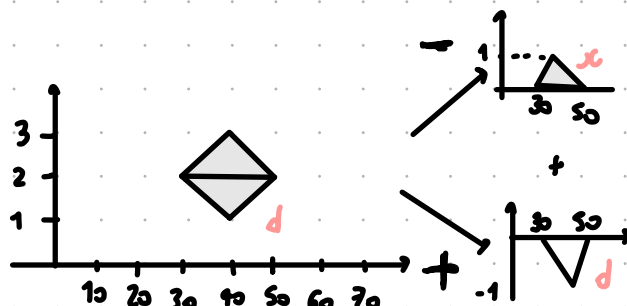
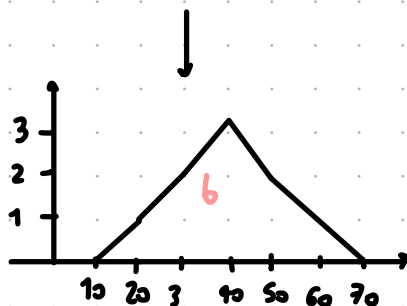
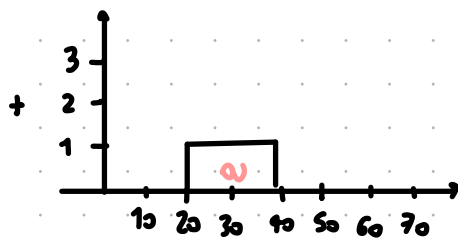
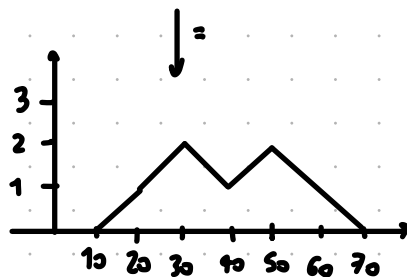
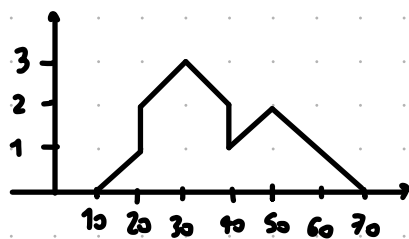
Calculate the energy of the portion of the signal between 0 and 40 μs .

Without performing the relevant calculation, can you estimate whether the energy of the remaining portion (from 40 μs onward) is greater or less than the energy in the previous tract?

How large is the coefficient a_0 of the Fourier series associated with the periodic signal $sp(t)$ that is obtained by replicating infinite times the part of $s(t)$ between between 0 and 80 μs ?

FOURIER TRANSFORM OF THE SIGNAL

- 1 - DIVIDE IN SIMPLE SIGNALS
- 2 - FIND $S(f)$ OF SIMPLE SIGNALS ← CAN BE SKIPPED
- 3 - COMPUTE FOURIER



$$w(t) = 1 \cdot \text{RECT}\left(\frac{t-30}{20}\right) \longrightarrow w(f) = 1 \cdot 20 \text{SINC}(f20) \cdot e^{-j2\pi f30}$$

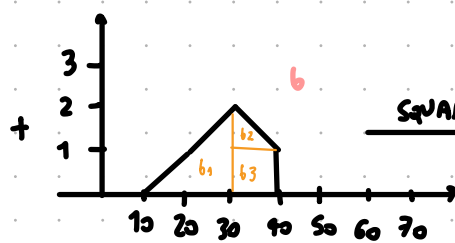
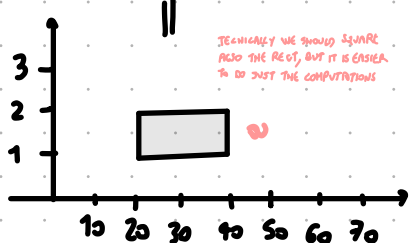
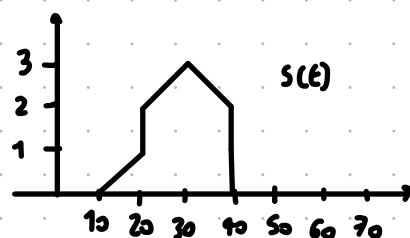
$$b(t) = 3 \cdot \text{TRIANG}\left(\frac{t-40}{60}\right) \longrightarrow b(f) = 3 \cdot \frac{60}{2} \text{SINC}^2\left(f \frac{60}{2}\right) \cdot e^{-j2\pi f40}$$

$$x(t) = 1 \cdot \text{TRIANG}\left(\frac{t-40}{20}\right) \longrightarrow x(f) = 1 \cdot \frac{20}{2} \text{SINC}^2\left(f \frac{20}{2}\right) \cdot e^{-j2\pi f40}$$

$$d(t) = -1 \cdot \text{TRIANG}\left(\frac{t-40}{20}\right) \longrightarrow d(f) = -1 \cdot \frac{20}{2} \text{SINC}^2\left(f \frac{20}{2}\right) \cdot e^{-j2\pi f40}$$

$$S(f) = w(f) + b(f) - x(f) + d(f)$$

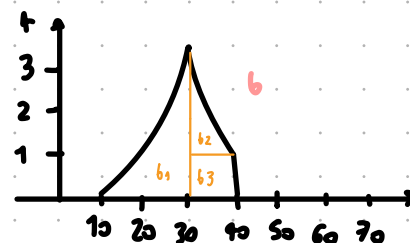
ENERGY OF SIGNAL BETWEEN 0 AND 40 μs



$$E = \int S(f)^2 df = E(w) + E(b) + 2R_{wb}(\phi)$$

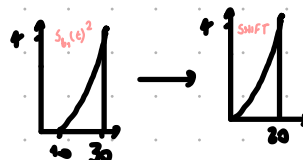
- DIVIDE $S(f)$ IN EASIER SIGNAL
- COMPUTE $S(f)^2$

- DO THE PRODUCT BETWEEN $S_w(f)$ AND $S_b(f)$
- COMPUTE AREA OF THE PRODUCT



$$E(w) = A^2 \text{BASE} = 1^2 \cdot 20 = 20$$

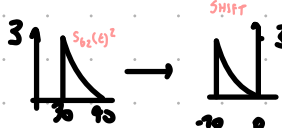
$$E(b) = \int S_b(f)^2 df = \int_0^{20} \frac{1}{100} f^2 df = \frac{1}{100} \cdot \frac{f^3}{3} \Big|_0^{20} \approx 26.6$$



$$y = \omega x^2$$

$$4 = \omega(20^2) \rightarrow \omega = \frac{1}{100}$$

$$E(b_2) = \int S_{b_2}(f)^2 df = \int_{-10}^0 \frac{3}{100} f^2 df = \frac{3}{100} \cdot \frac{f^3}{3} \Big|_{-10}^0 = \frac{3}{100} (0 - (-\frac{1000}{3})) = 10$$

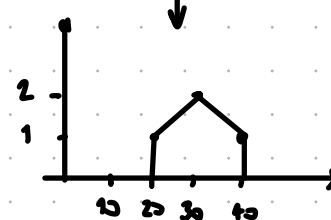
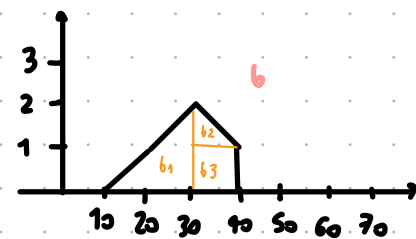
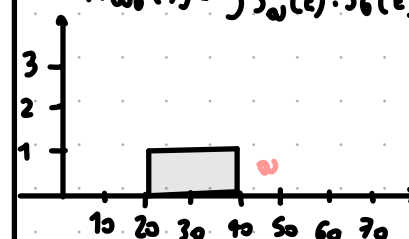


$$y = \omega x^2$$

$$3 = \omega(10)^2 \rightarrow \omega = 3/100$$

$$E(b_3) = A^2 \text{BASE} = 1 \cdot 10 = 10$$

$$R_{wb}(\phi) = \int S_w(f) \cdot S_b(f) df$$



$$R_{wb}(\phi) = \text{AREA} = 20 \cdot 1 + \frac{20 \cdot 1}{2} = 30$$

$$E = \int S(f)^2 df = E(w) + E(b) + 2R_{wb}(\phi)$$

$$= 20 + (26.6 + 10 + 10) + 2 \cdot 30$$